RUCHI SAHA

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PROFESSIONAL SUMMARY

Software Engineer with 5+ years of work experience, building high-scale distributed systems. Specialized in backend system design, microservices, high-availability systems and scalable architectures, with a burning passion in product and innovation.

SKILLS

Languages/Frameworks	Scala, Python, C#, Javascript, React, TypeScript, Java, C/C++
Cloud & Infrastructure	AWS, Azure, Kubernetes, Docker, CDN
Backend	Microservices, REST APIs, PostgreSQL, Redis, Large Scale Distributed Systems
AI / Machine Learning	LLMs, Semantic Search, Knowledge-Augmented Generation (KAG), LangChain, Ollama

WORK EXPERIENCE

Software Engineer 2, Microsoft India R&D Pvt Ltd (Apr'24 - Present)

- Stabilized a high-availability distributed system, serving 7B+ requests/day by reducing failure alerts by ~97% (900/day to 30/day) and failed requests (http 5xx) by ~99% (8.6M/min to 286/min) during peak traffic.
- Led cross-team service takeover and onboarding with just a 2 person team, authoring documentation, handling customer escalations and managing access/setup during high-ambiguity platform migration.
- Established standardized on-call, incident management, and deployment processes for the team, improving operational readiness and reducing production risk.
- Designed server-side rendering architecture and led security and privacy reviews - for Word, Excel, and PowerPoint home pages, to ensure compliance and robustness.
- Optimized performance of shared components in Microsoft apps (Word, Excel, PowerPoint, OneNote), improving boot/render time by 60%.

Backend Server Engineer, Samsung R&D Institute India-Delhi (Jan'21 - Apr'24)

- Redesigned a global microservices-based data ingestion pipeline powering Samsung TV+, processing 100GB+ daily, improving scalability, latency and increasing data correctness to more than 99%.
- Improved reliability of CDN-backed service handling 1M+ daily requests by reducing 5xx errors, improving system stability.
- Initiated and led department-wide hackathons and ideathons, which led to 5+ proof-of-concepts adopted for further development.

AI PROJECTS AND PATENTS

Knowledge-Augmented Generation (KAG) Assistant for Deep Search *(Patent Filed – Microsoft, Under Review)*

- Designed a novel semantic search algorithm for deep research across large knowledge bases
- Supports both intra-domain depth and cross-domain discovery in a single query (user-defined % across domains)
- Supports incremental knowledge updates; runs locally for private data use

Research Copilot (RAG-based AI for Science) *(Microsoft Global Hackathon – 3rd Place)*

- Built a RAG-based research assistant with over 2,000+ microbiology papers using GraphRAG
- Developed end-to-end system (backend + UI) using Ollama, Qwen 2.5, GraphRAG, and all-minilm (embeddings)

Agent SpongeHat (Personalized AI Work Agent)

- Developed a stateful AI agent that learns user and project context, enabling multi-persona reasoning
- Built end-to-end system (backend + UI + API) using FAISS, transformers (Hugging Face), GPT-4o, PyTorch

INNOVATION AND AWARDS

- Microsoft Global Hackathon 2024 — 3rd Place Winner
- Filed a patent for a novel Knowledge-Augmented Generation (KAG) algorithm enabling deep domain-specific and cross-domain research over private knowledge bases using LLMs.
- Served as a judge at Hacckaccino 4.0, a university hackathon held during the tech fest at Bennett University.
- Awarded Best in Development – Business Idea (UPMS) - Self-metadata generation to enhance search.
- 2× Spot Awards for engineering excellence.
- Won a prize for a chosen concept in the department-level ideathon

PUBLICATIONS AND RESEARCH

- Designed High-performance packet classification system for NIDS (FPGA Based), achieving ~5.9× performance and 5× resource efficiency (submitted to Computer Systems Science and Engineering)
- Proposed cryptography-based secure data deduplication for fog-to-cloud IoT systems, improving data integrity and storage efficiency (submitted to Computer Networks, Elsevier; Best Paper Award – CSAM IoT Security)

EDUCATION

Dual Degree (BTech + MTech) Computer Engineering, CGPA: 8.89/10
Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram